

Interview Management System — Feature Guide

A complete overview of what the system does, who uses each part, and how a candidate moves from "new application" to "hired or rejected."

Note on screenshots: Drop PNG/JPG images into `docs/screenshots/` using the filenames shown below each image placeholder, and they'll render inline when this doc is viewed on GitHub.

1. What this system is

A web-based hiring console that helps HR run the entire interview pipeline in one place. It replaces the usual spreadsheet + email back-and-forth with:

- A **single dashboard** for managing candidates, tests, interviews, and decisions.
- **AI assistance** for screening resumes, drafting coding problems, and scoring MCQ answers.
- **Automated emails** for every stage (invitations, reminders, shortlist, rejection).
- **Anti-cheat tools** baked into the candidate-facing test pages.
- A clean **interviewer portal** so external interviewers only see what they need.

2. Who uses it

Role	What they can do
HR / Admin	Full access — create candidates, manage questions, schedule interviews, make hire decisions, export data.
Interviewer	A separate login with access only to interviews assigned to them — review the candidate, submit ratings, and (if needed) request edits to a submitted review.
Candidate	No login. Receives email links to take tests or join interviews. Each link has a token + expiry.

3. The full candidate journey

This is the **happy path** from application to hire decision. Each step is covered in detail later in the doc.

1. HR creates candidate (uploads resume)
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2. AI scans the resume against a Job Description, returns a Match %
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3. HR approves or declines the resume
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4. HR sends the MCQ test link to the candidate
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5. Candidate takes the timed MCQ test (with photo + anti-cheat)
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6. AI auto-scores the answers – outcome: Shortlisted / Rejected / Disqualified
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7. (Optional) HR sends the Coding Test link
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8. Candidate solves coding problems in an in-browser code editor
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9. Backend auto-runs the test cases; HR rates the code 1-5 stars
↓
10. HR shortlists the candidate → schedules a Round 2 interview
↓
11. Interviewer conducts the meeting and submits a structured review
↓
12. HR reads the review and makes the final hire / culture-round decision

4. Module-by-module breakdown

4.1 Candidates

The home base for every applicant.

- **Create a candidate** — name, email, tech stack, experience level, number of MCQ questions, test duration, and an optional resume (PDF / DOC / DOCX up to 5 MB).
- **Resume scanning animation** — when a resume is uploaded, HR sees a live "document scanner" preview of the candidate's resume with a 4-step progress tracker (Create → Upload → Scan → Match) instead of a blank waiting screen.
- **Candidate list page** — searchable, filterable by status / experience, paginated.
 - **Inline quick-actions** per row: Approve / Decline (during resume review), Copy test link, Resend invite, Regenerate token, View details, Delete.

- **Export CSV** — downloads every candidate matching the current filters: name, email, tech stack, status, resume match %, coding test outcome, and more.
- **Candidate detail page** — the full profile, with:
 - Avatar, contact, tech-stack chips, experience level, status badge.
 - **Screening panel** — the AI's match %, green flags, red flags, JD title, and AI recommendation. (See §4.3.)
 - **Coding Test summary** card with a button that opens a dedicated review page.
 - **Interviewer review** (after Round 2 completes).
 - **Action bar** that adapts to the candidate's stage: Approve, Decline, Send test, Send coding test, Select for culture, Reject, Delete.

4.2 Job Descriptions (JD library)

A reusable bank of role descriptions, used by the AI to score resumes.

- HR adds a JD with: title, role, responsibilities, qualifications, nice-to-haves, and a min/max years-of-experience range.
- Each JD is **scoped to a tech stack** (e.g., a "React Developer" JD applies to candidates tagged `react`).
- JDs can be marked active / inactive without deletion.
- When a candidate is created with a matching tech stack, the system automatically picks the right JD for screening.

4.3 AI Resume Screening

Runs automatically when a resume is uploaded for a candidate whose tech stack has an active JD.

What it produces:

- A **Match %** (0–100) — how well the resume aligns with the JD.
- **Green flags** — specific things the resume gets right (technologies, years of experience, projects).
- **Red flags** — gaps or mismatches (missing skills, wrong experience range, etc.).
- A short **summary** in plain English.
- An **AI recommendation**: Approve or Decline.

HR can:

- **Approve** or **Decline** the resume. Both fire an email to the candidate.
- **Override the AI** — if HR disagrees, a confirmation modal appears warning that the AI recommendation is being overridden.

- **Re-screen** — if the JD changes or the candidate's tech stack is updated, HR can re-run the screening.

The AI uses a **fallback chain** for reliability: Gemini 2.5-flash → Gemini 2.5-flash-lite → Gemini 2.0-flash → Gemini 2.0-flash-lite → Groq llama-3.3-70b → Groq llama-3.1-8b. If all providers are down, screening is marked **failed** and HR reviews manually.

4.4 Question Bank

Used to generate the MCQ test.

- Each question has: prompt, options, correct answer, tech stack, difficulty (easy/medium/hard), and source (manual or AI-generated).
- HR can **draft questions with AI** by providing topic + difficulty.
- When HR creates a candidate, the system **samples questions** from the bank that match the candidate's tech stack — using a least-used-first strategy so questions rotate evenly.

4.5 MCQ Test (Round 1, the timed assessment)

The candidate receives an email with their test link.

What the candidate experiences:

1. Opens the link → **photo capture** step (selfie via webcam) to verify identity.
2. Test page loads with the timer running.
3. Questions appear one by one. Each has options.
4. **Anti-cheat:**
 - Tab switches are tracked and logged.
 - The page is locked into full-screen-like behavior.
 - **3 tab switches** → **auto-submitted as "Cheated"** (configurable threshold).
 - Photo from the start is stored as evidence.
5. When time runs out (or candidate clicks Submit), answers go to the server.

What happens on the backend:

- Answers are **auto-graded** against the bank's correct answers.
- A **percentage score** is computed.
- Based on score + cheat detection, the candidate's status flips automatically:
 - **≥ 60% and no cheat** → **ShortListed** (eligible for Round 2 or culture round).
 - **< 60%** → **Rejected**.
 - **Cheat detected** → **Cheated** (special status, no email).
- A **score report email** is sent to HR with the breakdown.

- A **result email** is sent to the candidate (different template per outcome).

If a coding test is also pending review, the MCQ auto-outcome is **suppressed** — the final decision waits until both tests are reviewed together.

4.6 Coding Test (optional second-round assessment)

Lets HR evaluate the candidate's coding ability with real programming problems, runnable in three languages.

HR side — Coding Problems library:

- HR builds a bank of coding problems: title, description (markdown), difficulty, tech stack, supported languages (JavaScript / Python / PHP), starter code per language, and test cases (input + expected output, can be marked hidden).
- Each problem can be **AI-drafted** in two ways:
 1. **Generate entire problem** from a topic — AI returns title, description, starter code, and test cases.
 2. **Generate starter code** only — AI scaffolds the stdin/stdout boilerplate for a language.
- Problems can be edited, deactivated, filtered by source (manual / AI), difficulty, language.

Sending the test:

- On the candidate detail page, HR clicks **Send coding test** → modal asks for: number of problems (1–5), duration (minutes), difficulty.
- System **samples problems** from the bank matching the candidate's tech stack. If the bank doesn't have enough matches, **AI generates new ones on the fly** and saves them to the bank for future use.
- An invite email goes to the candidate with the link.

Candidate side — the coding test page:

- IDE-style **split-pane layout** (HackerRank / LeetCode feel):
 - **Left pane:** Problem title, difficulty badge, full description (markdown rendered), example input/output cards in dark terminal style.
 - **Right pane:** VSCode-dark Monaco editor, language picker, starter code pre-loaded.
- **Big timer pill** at the top — turns red and pulses in the last minute. **Auto-submits** when time runs out.
- **Progress dots** at the bottom showing problem 1/3, 2/3, etc.
- **Anti-cheat:**
 - **Paste disabled** in the editor (with a toast warning).
 - **Copy disabled** at the page level.

- **Right-click / context menu disabled.**
- **Tab switches counted** — a pill at the top turns amber after 1 switch and red after 3+. A warning modal appears on return.
- **Tab switch count persists across page refresh** (stored in localStorage).
- Candidate writes solutions for each problem, navigates with Previous/Next, hits **Submit and finish** on the last one.

Auto-execution (server-side):

- On submit, the backend runs each problem's code through the **Piston public API** (a free multi-language code-execution sandbox).
- Each test case runs with the candidate's code as a child process; stdout / stderr / exit code are captured.
- A test case **passes** if `actualStdout.trim() === expectedStdout.trim()` and exit code is 0.
- **Hidden test cases** are run too — the candidate only sees the visible (sample) ones during the test.

HR review — the dedicated coding test page:

- Accessed via the **View coding test submission** button on the candidate detail page (separate route to keep the candidate detail clean).
- Top summary card: sent/submitted dates, outcome pill (Pending review / Shortlisted / Rejected), three stat boxes — test cases passed (with %), tab switches, languages used.
- For each problem:
 - Difficulty badge, language pill, pass-ratio pill.
 - **Test case results** — each case in a card with a green/red border, colored pass/fail icon, Input / Expected / Got blocks (mismatched "Got" shows dark red).
 - **Read-only Monaco editor** showing the candidate's code in VSCode-dark theme.
 - **Re-run tests** button (re-executes against Piston without saving a new submission).
 - **5-star rating** + comment box per problem.
- **Final decision panel** (only when all problems are rated): Shortlist or Reject buttons. Each fires the corresponding Round 1 email to the candidate.

4.7 Interviews (Round 2)

After Round 1 (MCQ + optional coding), HR schedules a live interview with a specific interviewer.

Interviewer management:

- HR adds interviewers with name, email, and **expertise areas** (e.g., `react`, `node`, `system design`).

- On creation, the interviewer gets an email with a **password setup link** — they pick their own password.
- Interviewers can be marked active / inactive.

Scheduling an interview:

- HR opens a shortlisted candidate, clicks **Schedule interview**, picks:
 - An interviewer (filtered by overlapping expertise).
 - Date + time, duration, meeting URL (Zoom / Meet / Teams link HR pastes in).
 - Optional notes for the interviewer.
- The system fires two emails simultaneously:
 - **To the candidate:** meeting time, link, interviewer name.
 - **To the interviewer:** candidate summary, meeting time, link, **a magic interviewer-only URL** to load the review form.

Interview reminders:

- A scheduler runs every minute on the backend.
- **30 minutes before** each scheduled interview, it emails BOTH the candidate and the interviewer with a reminder + meeting link.
- A `reminderSentAt` flag prevents double-sends, and it auto-resets if HR reschedules the interview.

Reschedule flow:

- Interviewers can request a new time + reason from their portal.
- HR sees a **"Pending reschedule" banner** on the interview detail page with: current time, proposed time, reason.
- HR can **Approve** (reschedule is applied, both parties emailed) or **Reject** (interview stays, interviewer notified). HR can add a decision note.
- Full reschedule history is visible per interview.

Interview detail page (HR view):

- Top card: schedule time, duration, status badge, candidate + interviewer info, meeting URL with copy button, magic link copy buttons, notes.
- **Action bar:** Edit, Mark complete, Cancel (with reason prompt).
- **Pending reschedule banner** when applicable.
- **Reschedule history** collapsible list.
- **Interviewer review** section (appears once the interview is completed).

Interview list page:

- Filter by status / from-date / to-date.
- **Export CSV** — downloads every interview with: time, candidate, interviewer, status, meeting URL, notes, completion details, etc.

4.8 Interviewer Portal

A separate dashboard so external interviewers only see their assigned interviews and nothing else.

- **My interviews list** — upcoming + past, with status.
- **Interview detail** — candidate name, resume download (if available), tech stack chips, scheduled time, notes from HR.
- **Review form** — three required 1–5 star ratings:
 - **Knowledge** — technical depth.
 - **Communication** — clarity, articulation.
 - **Confidence** — composure, problem-solving demeanor.
 - Plus a **free-text comments** box.
- **Auto-complete on submit** — submitting the review automatically marks the interview as **completed**, no separate "mark complete" step.
- **Edit-request flow** — if the interviewer wants to amend their review later, they submit an edit request with a reason. HR reviews + approves/rejects with a decision note.

4.9 Reviews & Edit Requests (HR oversight)

- All interviewer reviews appear on both the **candidate detail page** and the **interview detail page** with: interviewer name, average rating, per-axis stars, comments, submitted-at, edit count.
- **Edit-request center** at </admin/review-edit-requests> — a queue of pending requests across all interviews. HR can approve or reject each with a note.
- Full **edit history** is shown on the review panel — every previous request, decision, reason, and HR's decision note.

4.10 Submissions module

A dedicated browsing view of all MCQ submissions across candidates.

- Filterable, paginated list with score, outcome, cheat status.
 - Click into a submission to see: each question, the candidate's answer, the correct answer, per-question evaluation, and the photo captured at the start of the test.
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5. Automation summary

Things that happen without HR doing anything:

Trigger	What fires
Resume uploaded	AI screening starts, match % + flags stored, candidate detail updates.
Candidate created without resume	Status set to <code>resume_pending</code> (manual approve required).
Resume Approved	Email to candidate informing they're shortlisted for the test.
Resume Declined	Polite rejection email.
Send MCQ test	Email to candidate with timed test link.
MCQ submitted	Auto-graded, status flips, score report to HR, result email to candidate.
Tab switches exceed threshold	Test auto-submits, status set to <code>Cheated</code> .
Send coding test	Email with link, problems sampled (or AI-generated).
Coding test submitted	Test cases auto-run via Piston, HR notification email with pass/fail count.
Coding test Shortlisted	Round 1 shortlist email to candidate.
Coding test Rejected	Round 1 rejection email to candidate.
Interview scheduled	Two emails: candidate + interviewer.
30 min before interview	Reminder emails to candidate + interviewer.
Interview rescheduled	Re-fired notifications to both parties; reminder flag reset.
Reschedule approved/rejected	Notification to interviewer with HR's decision note.
Interviewer submits review	Interview auto-completed.
HR shortlists for culture / rejects	Final-stage email to candidate.

6. Anti-cheat measures

Both the MCQ test and the coding test have layered protections:

MCQ test:

- **Photo capture** before the test starts — stored as visual proof.
- **Tab-switch counter** — N switches → auto-submit with **Cheated** status.
- **Server-side timer enforcement** — even if a candidate fakes the client clock, the backend tracks the real elapsed time.
- **Single-use token** — once submitted, the link can't be reused.
- **Token expiry** — links expire 24 hours after issue.

Coding test:

- **Paste blocked** in the Monaco editor (toast warning shown).
- **Copy / Cut blocked** at the page level (Ctrl/Cmd + C/V/X intercepted).
- **Context menu disabled** (no right-click).
- **Tab-switch counter** with a visible amber/red pill that updates live; HR sees the final count on the review page.
- **Tab-switch count persists across refresh** (localStorage, keyed by token).
- **Auto-submit on timer expiry**.
- **Test cases run server-side only** — candidate code never executes in the browser.
- **Hidden test cases** that the candidate never sees, so the answers can't be hard-coded.

7. Reporting & data export

- **CSV export from the Candidates page** — every column needed for offline reporting (name, email, tech stack, status, match %, coding outcome, dates, links).
 - **CSV export from the Interviews page** — schedule, candidate, interviewer, status, meeting URL, completion details, cancel reasons.
 - Both exports **respect the current filters** — so if you've filtered "Status = Shortlisted", only those rows are exported.
 - Files open cleanly in Excel / Google Sheets (UTF-8 BOM + Excel-friendly escaping).
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8. Quick reference — where to find what

Need to...	Go to
Add a new candidate	Candidates → + New candidate
See AI's match % for a candidate	Candidates → click candidate → Screening panel
Approve / Decline a resume	Candidates list (inline) or candidate detail page
Send the MCQ test	Candidate detail page → Send test (after resume approved)
See an MCQ submission	Submissions → click row
Build the question bank	Questions → + New question or AI generate
Create / edit a JD	Job Descriptions → + New JD
Build a coding problem	Coding Problems → + New problem (manual or AI)
Send the coding test	Candidate detail page → Send coding test
Review coding submission	Candidate detail → View coding test submission
Add an interviewer	Interviewers → + New interviewer
Schedule an interview	Candidate detail (after shortlisted) → Schedule interview
Approve a reschedule request	Interview detail → Pending reschedule banner
See an interviewer's review	Candidate detail OR Interview detail
Approve a review edit request	Admin → Edit requests
Export everything	Candidates / Interviews → ↓ Export CSV

9. Tech stack (for IT / dev reference)

- **Backend:** Node.js, Express, MongoDB (Mongoose), Joi validation, Jest tests.
- **Frontend:** React, Redux Toolkit, Vite, SCSS, React Router, Monaco editor.
- **AI providers:** Google Gemini (primary, multiple model tiers) → Groq (fallback) for resume screening, MCQ grading, question drafting, problem drafting.
- **Code execution:** Piston public API (<https://emkc.org/api/v2/piston/execute>) — no Docker, no API key, supports JS / Python / PHP.

- **Email:** SMTP via Nodemailer.
- **File storage:** Cloudinary (resumes, photos).
- **Auth:** JWT with role claims (admin / interviewer), token-based public links for candidates.